ReAct

Why Do We Need ReAct?



- **Vanilla LLMs have fatal flaws**
  - Hallucination
    - LLMs guess when they don't know
    - E.g."The CEO of Acme Corp is John Smith" (made up)
  - No External Data
    - Can't access live information
    - Can't tell you today's weather or stock price
  - Single Pass
    - One chance to get it right
    - Can't adapt when initial approach fails
- Some tasks can't be solved in a single LLM call
- When a question requires looking up data, processing it, and making a decision based on the result, a one-shot response will either hallucinate the answer or give a shallow one

The ReAct Analogy



- **Think Like a Detective**
  - The detective doesn't just stare at the evidence and guess the answer
  - And they don't just randomly run around town looking for clues without thinking.
- **The Detective's Process:**
  - **Think** about what they know and need to find out
  - **Act** by looking up a record or visiting a location
  - **Observe** the new information
  - **Think** again about how this changes the theory
  - **Act** by interrogating a suspect
  - **Observe** the alibi
  - **Think** again... and finally conclude
- **ReAct does exactly this, but for AI agents!**

What is ReAct?



- **ReAct** (Reasoning + Acting) is an architectural pattern that interleaves **chain-of-thought reasoning** with **tool-calling actions** in a tight, synergistic feedback loop
- **Introduced by:** Yao et al., 2022 - "Synergizing Reasoning and Acting in Language Models"
- It is a behavior triggered by how you write your prompt, not by direct changes you make to the model
- This setup turns a basic chat into a structured process where the model explains its plan before it ever touches a tool

Step-by-Step

- **Thought:** AI analyzes the problem and plans next step
- **Action:** AI calls an external tool or API
- **Observation:** System feeds the tool result back
- **Repeat** until enough information is gathered
- **Final Answer:** Deliver the response

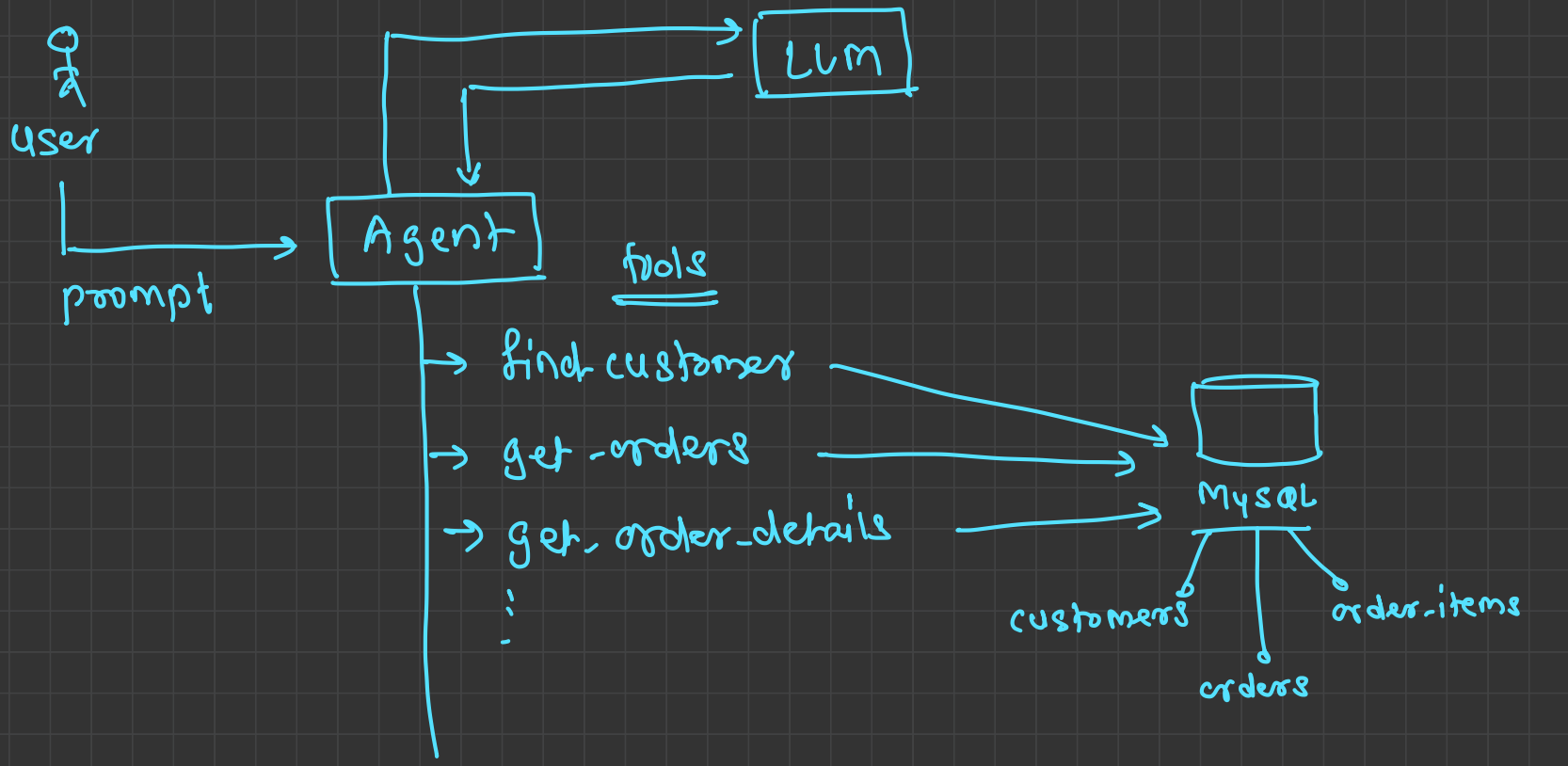


Detailed Example



- **User:** "What is the weather in Tokyo right now, and should I pack an umbrella?"
- **Thought 1:** I need the current weather in Tokyo. I will call the weather API.
- **Action 1:** get_weather(city="Tokyo")
- **Observation 1:** "Current conditions: 22°C, Overcast, 80% humidity, light rain expected in 2 hours."
- **Thought 2:** Rain is expected in 2 hours. I should advise packing an umbrella. Also check if a jacket is needed.
- **Action 2:** compare_temp(22°C)
- **Observation 2:** "22°C is mild. Jacket not required."
- **Thought 3:** I have all the information needed.
- **Final Answer:** "It is currently 22°C and overcast in Tokyo. Light rain is expected in about 2 hours, so yes, you should pack an umbrella. A jacket is not necessary for this temperature."

Customer support Agent



Structural Components

- Reasoning Engine (LLM)
  - Interprets prompts, creates plans
  - Generates Thought-Action traces
- Tool Layer (External Functions/APIs)
  - Search engines, databases, APIs
  - Executes actions and returns observations
- Working Memory (Context Window)
  - Stores thoughts, actions, observations
  - Maintains conversation history
- Scratchpad (Reasoning Trace)
  - Visible step-by-step logic
  - Auditable decision trail
- Orchestrator (Control Loop)
  - Manages Think-Act-Observe cycle
  - Passes outputs to tools, feeds results back

Advantages of ReAct



- **Enhanced Reasoning**
  - Transparent thinking process
  - Better decisions through thoughtful consideration
  - Error recognition and correction
- **Improved Accuracy**
  - Grounded in external facts, not hallucinations
  - Verification loops for double-checking
  - Adaptive strategies based on observations
- **Debuggability**
  - Reasoning traces show why decisions were made
  - Step-by-step inspection
  - Failure analysis capabilities
- **Dynamic Adaptation**
  - Context-aware adjustments
  - Flexible, non-predetermined sequences
  - Recovers from errors with alternative approaches

Limitations



- **Increased Latency**
  - Issue: Multiple LLM calls add time
  - Impact: 3-10x slower than single-shot responses
  - Each cycle requires a separate forward pass
- **Higher Token Costs**
  - Issue: Reasoning traces consume tokens without direct output
  - Impact: 2-5x more tokens than direct tool use
  - One unbounded query can cascade into 50-500 LLM calls costing $5-$50
- **Potential for Infinite Loops**
  - Agent can get stuck in unproductive cycles
  - Example: "Page not found" → "I'll search again" → "Page not found"...
  - Must implement maximum iteration limits
- **Context Window Limitations**
  - All thoughts, actions, observations must fit in context
  - Complex tasks with many steps can exceed token limits
  - Risk of context loss and inaccurate results

When to use ReAct?



- **Interactive tasks** where the goal isn't well-defined upfront
- **Dynamic user queries** that require adaptation (e.g., document chat, follow-up questions)
- **Exploratory data analysis** - "What patterns do you see in this report?"
- **Zero-shot reasoning** with unpredictable inputs
- **Need to learn** what tools your agent actually requires
- **Debuggability** and auditability are important

When to avoid?



- Control flow is known at design time (use a workflow instead)
- Latency SLOs are strict
- Short-horizon tasks (fewer than 10 steps)
- Domain requires auditable deterministic branches
- Compliance and auditability must be strictly enforced